Cubli: study guide and first math problem

1. What the ETH reference says

Michael Muehlebach (2018), *On the Stabilization of Unstable Systems - Some Comments and Applications*, ETH thesis no. 24789. Local PDF | Official record.

The thesis has four parts: Cubli, a ducted-fan flying vehicle, constrained LQR/MPC approximations, and distributed state estimation. **Read Part A first**; the other parts are not prerequisites for building a balancing cube. Page numbers below are printed thesis pages; PDF viewer pages are also given.

Read	Main result	What it means for this project
P1, dynamics: pp. 25-28 (PDF 44-47)	A 150 mm cube with three perpendicular reaction wheels; gravity, inertia, and angular momentum determine its motion.	Motor torque accelerates a wheel and reacts on the frame. Wheel speeds are states, not just outputs.
P1, nonlinear control: pp. 28-37 (PDF 47-56)	Backstepping with a Lyapunov argument stabilizes the upright set almost everywhere in the ideal model. Feedback linearization supports trajectory tracking.	First understand stability proofs and the model; the result is not a guarantee under arbitrary motor limits or impacts.
P1, jump-up: pp. 37-45 (PDF 56-64)	Braking transfers wheel momentum to the cube. A gradient-based learning method adjusts initial wheel speeds; a guiding controller tracks the rise.	Jump-up is a separate hybrid maneuver, not ordinary balancing with larger gains.
P1, experiments: pp. 45-47 (PDF 64-66)	Reported steady-state RMS inclination error below 0.025 degrees; recovery in under 1.8 s after a 0.17 N m, 60 ms disturbance on one wheel.	These are results for ETH's apparatus, not performance promises for a printed prototype.

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P2, estimation: pp. 51-78 (PDF 70-97)	Multiple accelerometers and a non-accelerated pivot assumption estimate tilt using rigid-body kinematics, without mass/inertia knowledge.	A moving accelerometer measures rotational acceleration as well as gravity; simply treating its reading as gravity can give wrong tilt.

Important limitations: the reduced gravity-vector representation describes tilt but discards yaw orientation. In the ideal frictionless-pivot model, angular momentum about gravity is conserved: one cannot generally force both frame yaw rate and all wheel speeds to zero. Ground friction changes this. ETH's almost-global balancing result concerns an upright set, not arbitrary full-attitude regulation with every wheel stopped.

INDI is not the controller developed in these Cubli chapters. It is our proposed comparison. Start with their dynamics and a conventional baseline; do not assume INDI will outperform it without experiments.

2. Nonlinear-control book

Hassan K. Khalil, *Nonlinear Systems*, 3rd edition, Prentice Hall, 2002, ISBN 9780130673893. Author's companion site | Verified contents | Publisher. Obtain the book through a library or publisher; only its citation and links are included here, not a copied textbook.

Suggested order: section 1.2.1 (pendulum), chapter 2 (phase portraits), sections 4.1-4.3 (Lyapunov, invariance, linearization), section 12.2 (stabilization via linearization), chapter 13 (feedback linearization), then section 14.3 (backstepping). Chapters 4 and 13 are the priority.

3. Similar concept: a reaction-wheel inverted pendulum

Think of one Cubli face on a fixed horizontal pivot, carrying one motor and wheel. This has the same gravity-versus-reaction-torque mechanism, without three-dimensional coupling. The following is an original educational exercise, not a reproduction of a textbook problem or ETH's full model.

Assume a rigid body, fixed frictionless pivot, ideal motor torque, no impacts, and no wheel/body drag. Let:

- θ: body angle from upright; $\omega = \dot{\theta}$.
- ϕ: wheel angle relative to the body; $s = \dot{\theta} + \dot{\phi}$ is its absolute speed.
- I: pivot inertia of the non-spinning assembly, including wheel-center translation but excluding the wheel's axial spin inertia J.

- m: total mass; l: pivot-to-total-center-of-mass distance.
- τ: torque on the wheel; the reaction torque on the body is $-\tau$.

Use these **illustrative parameters, not a hardware shopping specification**:

$$m = 0.40\,\text{kg}, \quad l = 0.08\,\text{m}, \quad I = 0.004\,\text{kg}\,\text{m}^2, \quad J = 0.0002\,\text{kg}\,\text{m}^2, \quad g = 9.81\,\text{m/s}^2.$$

Problems to solve

1. Start with $L = \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}J(\dot{\theta}+\dot{\phi})^2 - mgl\cos\theta$. The generalized forces for (θ, ϕ) are $(0, \tau)$ because the motor is internal. Derive the two Euler-Lagrange equations and show $I\dot{\omega} = mgl\sin\theta - \tau$ and $J\dot{s} = \tau$.
2. Linearize at $(\theta, \omega, s, \tau) = (0, 0, 0, 0)$. Write A, B for $x = (\theta, \omega, s)$; show the upright angle is open-loop unstable.
3. For the **angle subsystem**, choose $\tau = mgl\sin\theta + I(k_p\theta + k_d\omega)$. Find the closed-loop equation and choose gains for a critically damped response with natural frequency 8 rad/s.
4. Use $V = \frac{1}{2}k_p\theta^2 + \frac{1}{2}\omega^2$ to prove angle convergence. Explain why $\dot{V} \leq 0$ alone is insufficient and use LaSalle's principle. Does this also prove $s \to 0$?
5. Starting from $\theta = 5$ degrees, $\omega = s = 0$, calculate the initial torque. Simulate the nonlinear plant with $|\tau| \leq 0.20$ N m and a wheel-speed ceiling $|s| \leq 300$ rad/s. Detect and stop/report a speed-limit violation; do not clip s, since that would introduce an unmodeled braking torque.
6. Write $\dot{\omega} = f(\theta) + G\tau$ and derive an INDI update for target acceleration $v = -k_p\theta - k_d\omega$, using a recent measured acceleration α_0 and the corresponding applied torque τ_0. Explain what fails if the acceleration measurement is delayed or noisy.

Checkpoints: open after attempting the problems

- For the Euler-Lagrange coordinates, $q_1 = \theta$ (body angle) and $q_2 = \phi$ (wheel angle relative to the body). Their generalized forces are $Q_1 = Q_\theta = 0$ and $Q_2 = Q_\phi = \tau$: the motor applies $-\tau$ to the body and $+\tau$ to the wheel, so its virtual work is $-\tau\,\delta\theta + \tau\,\delta(\theta+\phi) = \tau\,\delta\phi$. Since $\partial L/\partial\dot{\theta} = I\dot{\theta} + J(\dot{\theta}+\dot{\phi})$ and $\partial L/\partial\theta = mgl\sin\theta$, the θ equation is $(I+J)\ddot{\theta} + J\ddot{\phi} - mgl\sin\theta = 0$, or $(I+J)\ddot{\theta} + J\ddot{\phi} = mgl\sin\theta$. The ϕ equation is $J(\ddot{\theta}+\ddot{\phi}) = \tau$, i.e. $J\dot{s} = \tau$. Substituting this into the θ equation gives $I\ddot{\theta} = mgl\sin\theta - \tau$, or $I\dot{\omega} = mgl\sin\theta - \tau$.
- The coupled equations are $(I+J)\ddot{\theta} + J\ddot{\phi} = mgl\sin\theta$ and $J(\ddot{\theta}+\ddot{\phi}) = \tau$.
- With $a = mgl/I = 78.48\ \text{s}^{-2}$: $A = \begin{bmatrix} 0 & 1 & 0 \\ a & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ -1/I \\ 1/J \end{bmatrix}$. The open-loop eigenvalues are 0 and approximately $\pm 8.86\ \text{s}^{-1}$.
- Ideal cancellation gives $\ddot{\theta} + k_d\dot{\theta} + k_p\theta = 0$; choose $k_p = 64\ \text{s}^{-2}$, $k_d = 16\ \text{s}^{-1}$.

- $\dot{V} = -k_d\omega^2$. The largest invariant set with $\omega = 0$ also has $\theta = 0$, proving angle convergence in the ideal, unsaturated angle model. This is not a global physical Cubli proof.
- Wheel speed need not converge to zero: $s(t) = s(0) + J^{-1}\int_0^t \tau\, dt$. The angle-only controller leaves the wheel's final speed unconstrained. As an extension, use full-state LQR to include wheel-speed regulation.
- The initial torque is approximately 0.0497 N m. Even when initial torque is feasible, check the entire trajectory's limits.
- $f = (mgl/I)\sin\theta$, $G = -1/I$. Locally, $\dot{\omega} \approx \alpha_0 + G(\tau - \tau_0)$, hence $\tau = \tau_0 + I(\alpha_0 - v)$. This assumes gravity changes little over the increment and applied torque responds fast enough. Time-align/filter the torque and acceleration signals consistently and include actuator dynamics and saturation.